STM32 X-CUBE-AI Integration Steps

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Step-by-Step Process for STM32 X-CUBE-AI Project Integration

1. **Create a New Project in STM32CubeMX**
   - Open STM32CubeMX.
   - Create a new project for your STM32 device (e.g., STM32F401RE).
   - Configure the necessary peripherals for the project, such as GPIO, USART for debug printing, and any other peripherals you need for the application.
   - Under the *Middleware* section, enable **X-CUBE-AI**.
   - Set up the clock configuration and ensure the clock speed is set to the maximum allowable for your device.
2. **Add X-CUBE-AI Component**
   - In the *Pinout & Configuration* tab, go to **Software Packs** → **X-CUBE-AI**.
   - Enable **AI Libraries**. This will include the necessary libraries for the AI framework.
3. **Import the Neural Network Model**
   - Click on the **X-CUBE-AI** button on the toolbar.
   - Click on **New Network** and choose the desired network name (e.g., `network`).
   - Click **Import** and browse to your Keras `.h5` or `.keras` model file that you want to integrate.
   - After importing the model, configure the parameters such as the number of layers, nodes, and activations as necessary.
   - Click on **Analyze** to ensure the model is compatible and fits within the memory constraints of your device.
   - If successful, you should see a green checkmark.
4. **Generate the Code**
   - Once the model analysis is complete, click on **Generate Code**.
   - Select the appropriate toolchain for code generation (e.g., STM32CubeIDE).
   - After code generation, open the project in the STM32CubeIDE.
5. **Modify the Generated Code as Needed**
   - Go to the `app_x-cube-ai.c` file and locate the functions `MX_X_CUBE_AI_Init()` and `MX_X_CUBE_AI_Process()`.
   - Add any necessary debug printing or modifications in the *USER CODE* sections.
   - Ensure the UART is configured correctly for serial output to monitor the model's output.
6. **Configure and Build the Project**
   - Build the project in STM32CubeIDE.
   - Ensure there are no build errors related to missing files or dependencies.
7. **Set Up Validation on Target**

- In the **X-CUBE-AI** GUI, set up **Validate on Target** by selecting the appropriate communication settings for your device.
- Select the USART that is connected to the ST-LINK's Virtual COM port.
- Enable the checkbox for **Automatic Compilation and Download**.
- Click on **Validate on Target**.

8. **Generate the AI Validation Project**
   - A new temporary project will be generated, compiled, and flashed to the device.
   - This temporary project will run the validation on the device and send the results back to the host PC.
   - Review the validation results to confirm if the model behaves as expected on the device.

9. **Integrate the Model in Your Application**
   - Once validated, you can integrate the generated network code (found in `network.c` and `network_data.c`) into your main application.
   - Use the `ai_network_run()` API to run inference with your inputs and outputs.